

# Zhao Shizhen

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Computer Science | Quantitative Finance | AI/Data Solutions | Backend Systems | Client-Facing Technical Delivery

## EDUCATION

### National University of Singapore

Bachelor's Degree in Computer Science w/ Double Major in Quantitative Finance (GPA 4.85/5.0)

Singapore

2025 – 2029

**Achievements:** Dean's List AY25/26 Semester 2, University Project Showcase 2026 Top 2 Winner

### Relevant Coursework:

Data Structures and Algorithms, Object Oriented Programming (OOP), Discrete Structures, Linear Algebra, Calculus II

## WORK EXPERIENCE

### KPLER

*Forward Deployed Engineer Intern, Commodities*

May 2026 - Present

- Designed data solutions over commodity and maritime datasets, translating client needs into technical deliverables.
- Built Python/SQL backend tools for historical extractions, data processing, and delivery
- Partnered with Product and Sales through the end-to-end sales cycle, owning technical scoping and customized demos.

### SOURCE ACADEMY

*DevOps ML Engineer, Internal Learning Platform*

Jan 2026 - May 2026

- Built LLM AI chatbot with internal knowledge base and Retrieval-Augmented Generation (RAG) techniques.
- Deployed models on AWS Lambda and Docker for scalable inference.
- Improved inference speed and accuracy by over 50% by implementing efficient chunking methods.

### SHOPEE

*Product Manager Intern, Internal HR Systems*

Sep 2025 - Dec 2025

- Designed agentic AI workflows using prompt engineering and tool integration to streamline HR processes.
- Integrated LLMs with APIs, databases, and internal systems to automate workflows, reducing processing time by 90%.
- Coordinated with development team to schedule requests within pipelines to balance user needs and engineer bandwidth.

## PROJECTS

### O(Alpha) - Quant Research & Agentic Trading Platform

GitHub: <https://github.com/shiverin/O-Alpha>

- Built a Go/Next.js platform with TimescaleDB, Redis, Docker and dashboards for portfolio research and paper trading.
- Implemented validation and paper-trading loops with walk-forward testing, cost stress checks and portfolio snapshots.

### Messenger - Real-Time Full-Stack Messaging App

GitHub: <https://github.com/shiverin/messenger>

- Built a private chat app with Django Channels, WebSockets, PostgreSQL, custom SPA-style UI, secure auth, and storage.

### Traffic Sign Classification with CNN

GitHub: <https://github.com/shiverin/CNNTraffic>

- Achieved 98.97% accuracy on the GTSRB dataset by designing and optimizing CNN architectures using TensorFlow.
- Implemented models with multiple convolutional pooling layers, tuned depth and dropout parameters to generalize.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Go, TypeScript, JavaScript, Java, C, R

**Backend/API:** REST APIs, Gin, FastAPI, Django, Flask, WebSockets, PostgreSQL/TimescaleDB, Redis

**Data/Analytics:** Snowflake, AWS Athena, ETL/data workflows, Pandas, Scikit-learn, Tableau, R Studio

**AI/ML:** LLMs, RAG, TensorFlow, PyTorch, YOLOv8, Whisper

**Cloud/Dev Tools:** AWS Lambda, Docker, Docker Compose, Git, Postman, Render, Supabase

## HACKATHONS

### BrainHack by DSTA – Finalist (Top 4)

Jun 2025

- Fine-tuned a Whisper model on 4,500 audio samples, achieving 0.98% word error rate (WER).
- Trained YOLOv8 on a dataset of 10,000 labelled images of military vehicles, achieving 67% test accuracy.
- Built a ResNet18 + CRNN model trained on 6,000 scanned text samples, achieving 99% character-level accuracy.
- Implemented a single-policy multi-agent DQN in a custom Scout-vs-Guards environment using PettingZoo.

## INTERESTS & HOBBIES

- Classical musician (piano & violin), Orchestral Performance, Leetcode, Watch Collecting, Chinese Chess